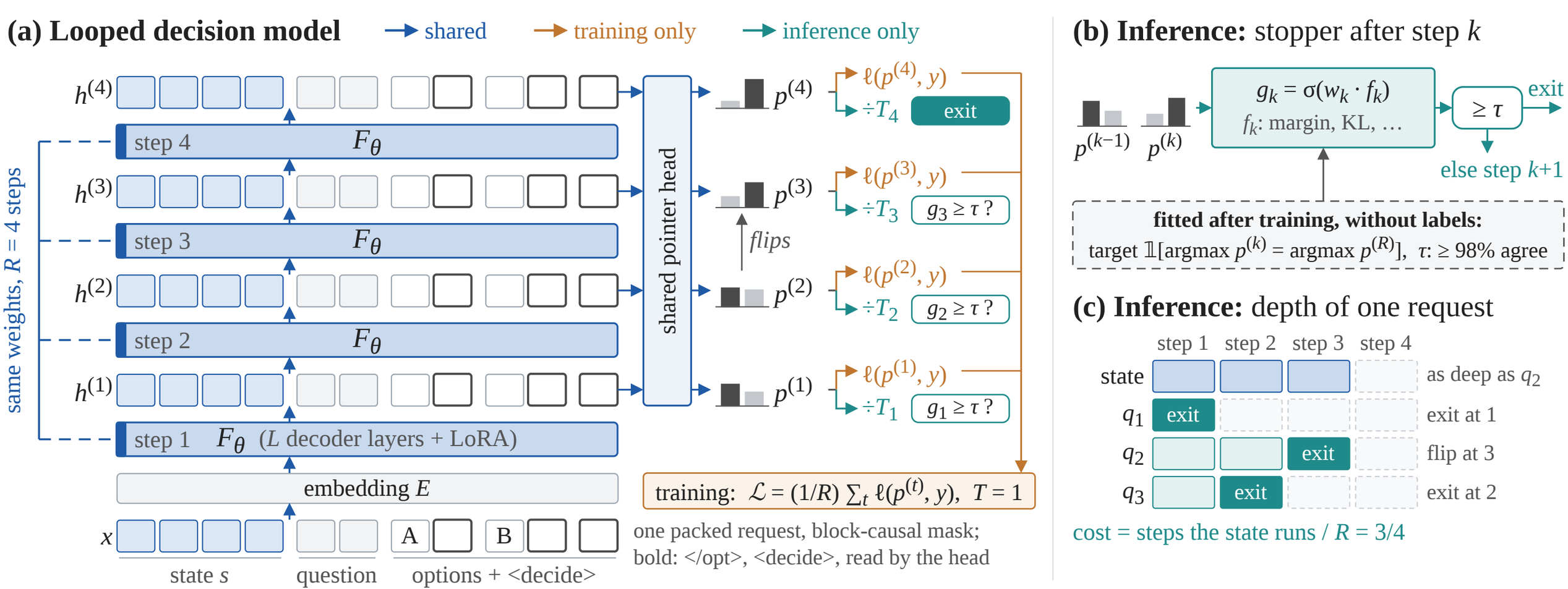

(a) Looped decision model
shared
training only
inference only
same weights, R = 4 steps
h(4)
step 4 F_θ
h(3)
step 3 F_θ
h(2)
step 2 F_θ
h(1)
step 1 F_θ (L decoder layers + LoRA)
embedding E
x
A
B
state s
question
options + <decide>
shared pointer head
p(4)
p(3)
flips
p(2)
p(1)
ℓ(p(4), y)
÷T4
exit
ℓ(p(3), y)
÷T3
g3 ≥ τ ?
ℓ(p(2), y)
÷T2
g2 ≥ τ ?
ℓ(p(1), y)
÷T1
g1 ≥ τ ?
training: ℒ = (1/R) ∑t ℓ(p(t), y), T = 1
one packed request, block-causal mask;
bold: </opt>, <decide>, read by the head
(b) Inference: stopper after step k
p(k−1)
p(k)
gk = σ(wk · fk)
fk: margin, KL, …
≥ τ
exit
else step k+1
fitted after training, without labels:
target 𝟙[argmax p(k) = argmax p(R)], τ: ≥ 98% agree
(c) Inference: depth of one request
step 1
step 2
step 3
step 4
state
as deep as q2
q1
exit
exit at 1
q2
exit
flip at 3
q3
exit
exit at 2
cost = steps the state runs / R = 3/4